YAKEEN

NEET 2.0 2027

Physics | Lecture - 8

Motion in a Straight Line

By- Manish Raj (MR Sir)



Todays Goal

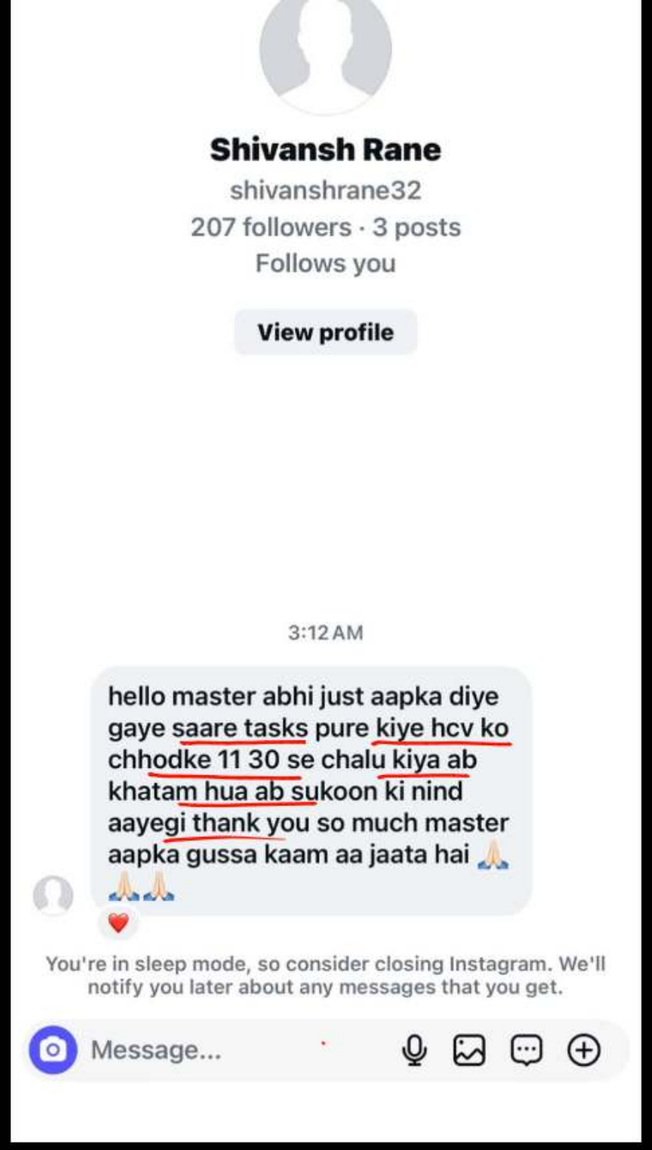
PhD on constant acc^{n}

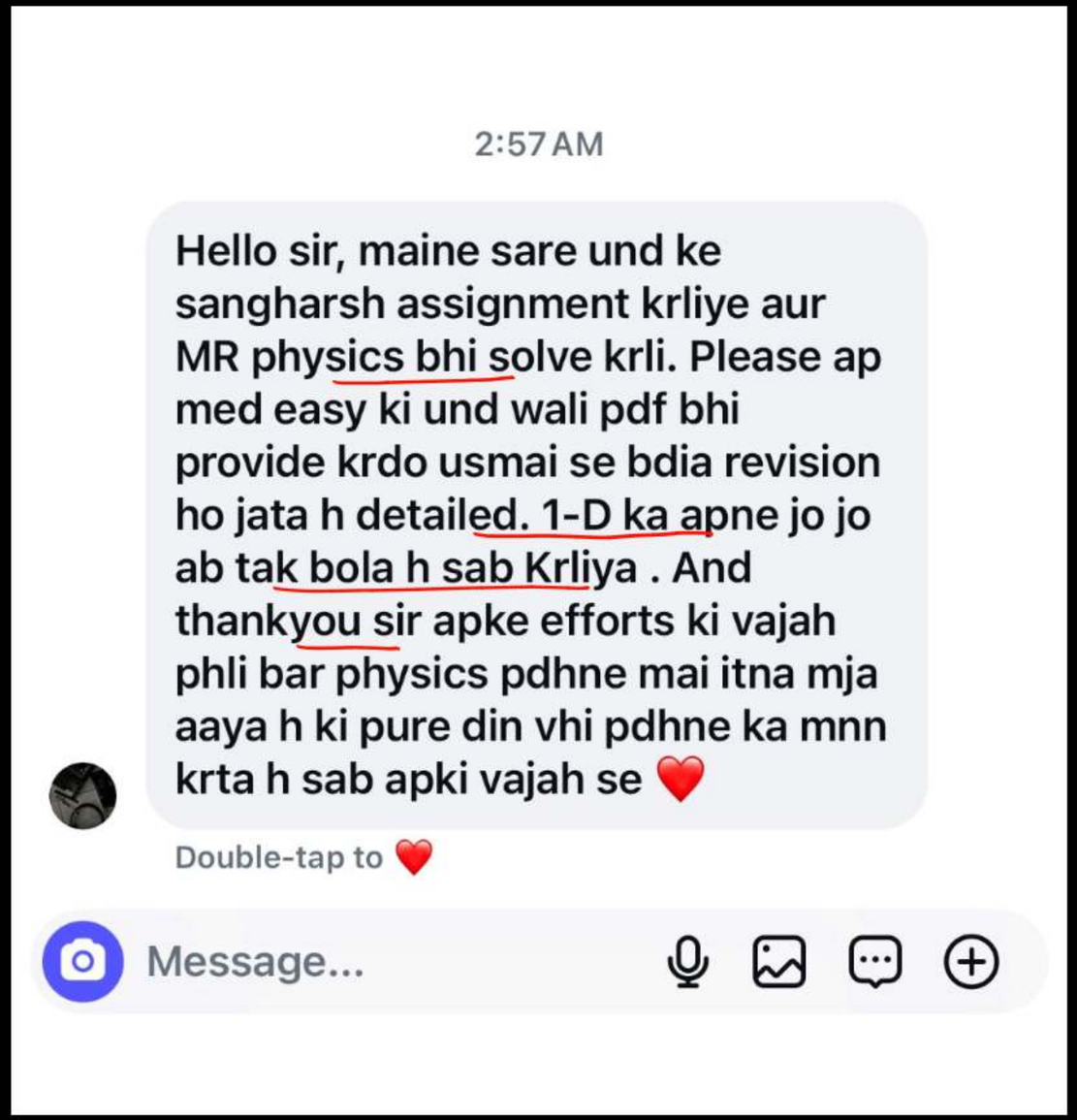
→ can we apply integ^{n} & diff^{n} → yes

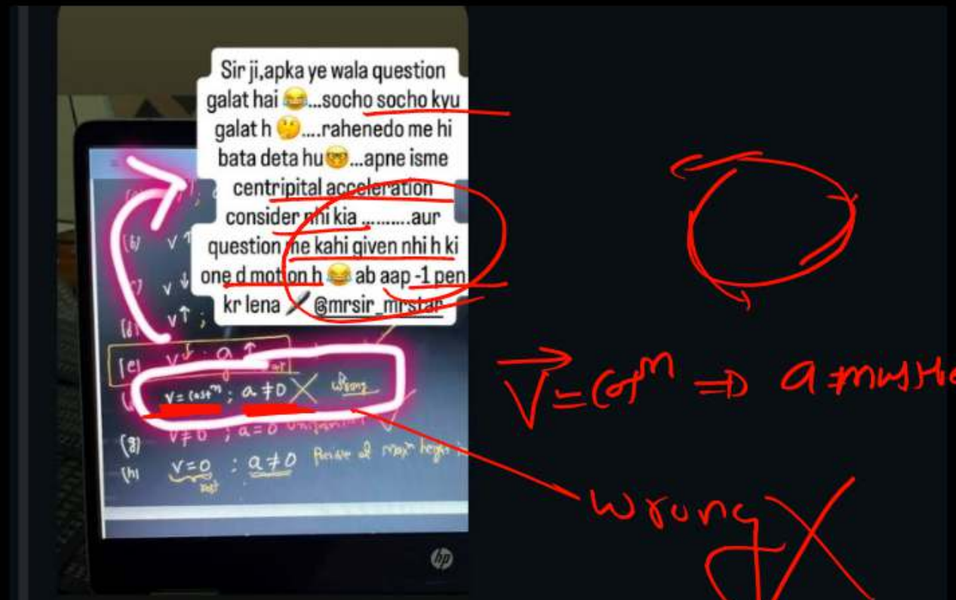
→ we generally uses eq^{n} of motion → yes



Join it

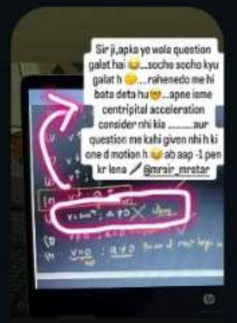

Shivansh Rane
shivanshrane32
207 followers · 3 posts
Follows you
View profile
3:12 AM
hello master abhi just aapka diye gaye saare tasks pure kiye hcv ko chhodke 11 30 se chalu kiya ab khatam hua ab sukoon ki nind aayegi thank you so much master aapka gussa kaam aa jaata hai 🙏🙏🙏
❤️
You're in sleep mode, so consider closing Instagram. We'll notify you later about any messages that you get.
Message...

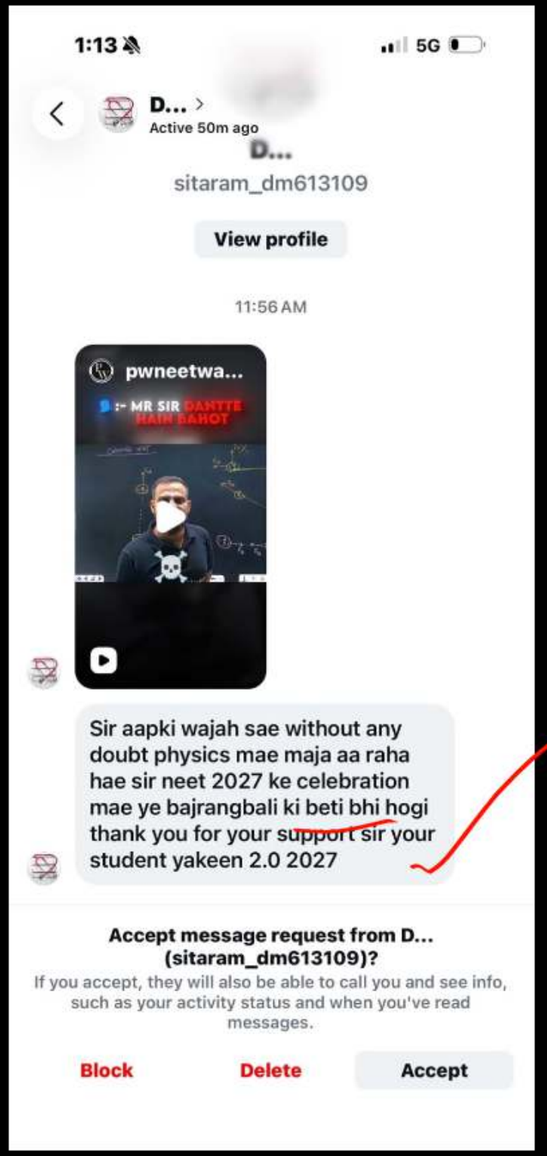

2:57 AM
Hello sir, maine sare und ke sangharsh assignment krliye aur MR physics bhi solve krli. Please ap med easy ki und wali pdf bhi provide krdo usmai se bdia revision ho jata h detailed. 1-D ka apne jo jo ab tak bola h sab Krliya . And thankyou sir apke efforts ki vajah phli bar physics pdhne mai itna mja aaya h ki pure din vhi pdhne ka mnn krta h sab apki vajah se ❤️
Double-tap to ❤️
Message...

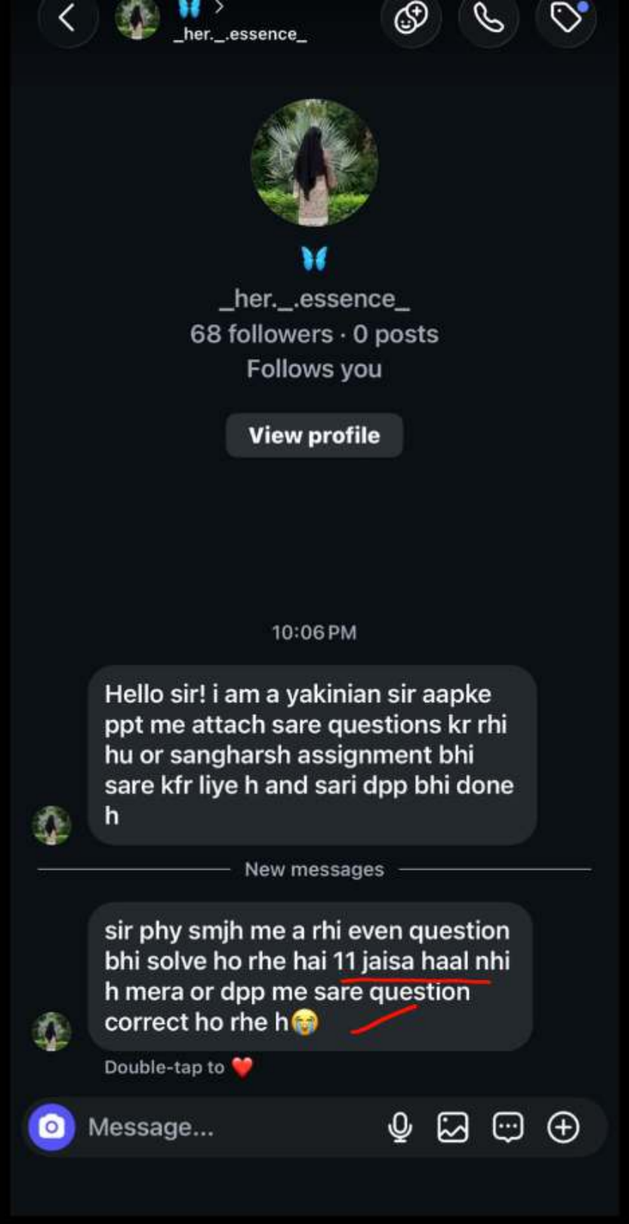


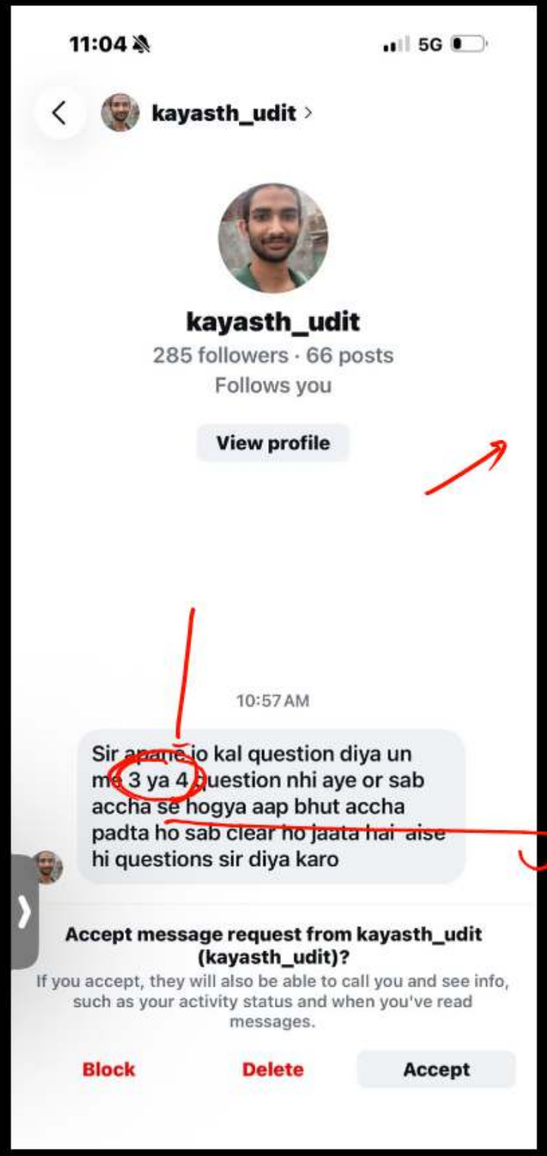
9:07 PM

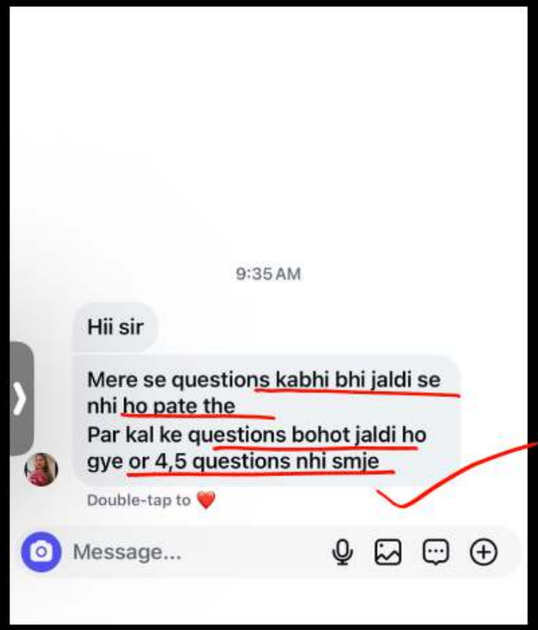
You replied to their story






her..essence_
her..essence_
68 followers · 0 posts
Follows you
View profile
10:06 PM
Hello sir! i am a yakinian sir aapke ppt me attach sare questions kr rhi hu or sangharsh assignment bhi sare kfr liye h and sari dpp bhi done h
New messages
sir phy smjh me a rhi even question bhi solve ho rhe hai 11 jaisa haal nhi h mera or dpp me sare question correct ho rhe h😭
Double-tap to ❤️
Message...


11:04
5G
kayasth_udit
kayasth_udit
285 followers · 66 posts
Follows you
View profile
10:57 AM
Sir apane jo kal question diya un me 3 ya 4 question nhi aye or sab accha se hogya aap bhut accha padta ho sab clear ho jaata hai aise hi questions sir diya karo
Accept message request from kayasth_udit (kayasth_udit)?
If you accept, they will also be able to call you and see info, such as your activity status and when you've read messages.
Block
Delete
Accept


9:35 AM
Hii sir
Mere se questions kabhi bhi jaldi se nhi ho pate the
Par kal ke questions bohot jaldi ho gye or 4,5 questions nhi smje
Double-tap to ❤️
Message...


7:11
Shiva
shivansh
2:57 AM
Hello sir, maine sare und ke sangharsh assignment krliye aur MR physics bhi solve krli. Please ap med easy ki und wali pdf bhi provide krdo usmai se bdia revision ho jata h detailed. 1-D ka apne jo jo ab tak bola h sab Krliya . And thankyou sir apke efforts ki vajah phli bar physics pdhne mai itna mja aaya h ki pure din vhi pdhne ka mnn krta h sab apki vajah se ❤️
Double-tap to ❤️
Message...
You're in sleep mode, notify you later a
Message...

GPT

(04) object starts his motion from rest ($u=0$) & constant acceleration $a = 10\,m/s^2$ then find

displacement in

$S_{1s} = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(10)(1)^2 = 5m$

$S_{2s} = \frac{1}{2}(10)(2)^2 = 5\times 4 = 20m$

$S_{3s} = \frac{1}{2}(10)(3)^2 = 5\times 9 = 45$

$S_{4s} = \frac{1}{2}(10)(4)^2 = 5\times 16 = 80$

$S_{5s} = \frac{1}{2}(10)(5)^2 = 5\times 25 = 125$

$S_{1s} : S_{2s} : S_{3s} : S_{4s} : S_{5s} = 5 : 20 : 45 : 80 : 125$

$= 1 : 4 : 9 : 16 : 25 : 36$

$= x : 4x : 9x : 16x : 25x$

$u=0$, $a=10m/s^2$

disp^m in n^{th} sec.

$S_{1^{st}} = u + \frac{a}{2}(2n-1) = 0 + \frac{10}{2}(2\times 1-1) = 5m$

$S_{2^{nd}} = \frac{a}{2}(2n-1) = \frac{10}{2}(4-1) = 5\times 3 = 15$

$S_{3^{rd}} = \frac{10}{2}(3\times 2-1) = 5\times 5 = 25$

$S_{4^{th}} = \frac{10}{2}(4\times 2-1) = 5\times 7 = 35$

$S_{5^{th}} = \frac{10}{2}(2\times 5-1) = 5\times 9 = 45$

$S_{1^{st}} : S_{2^{nd}} : S_{3^{rd}} : S_{4^{th}} : S_{5^{th}} = 5 : 15 : 25 : 35 : 45$

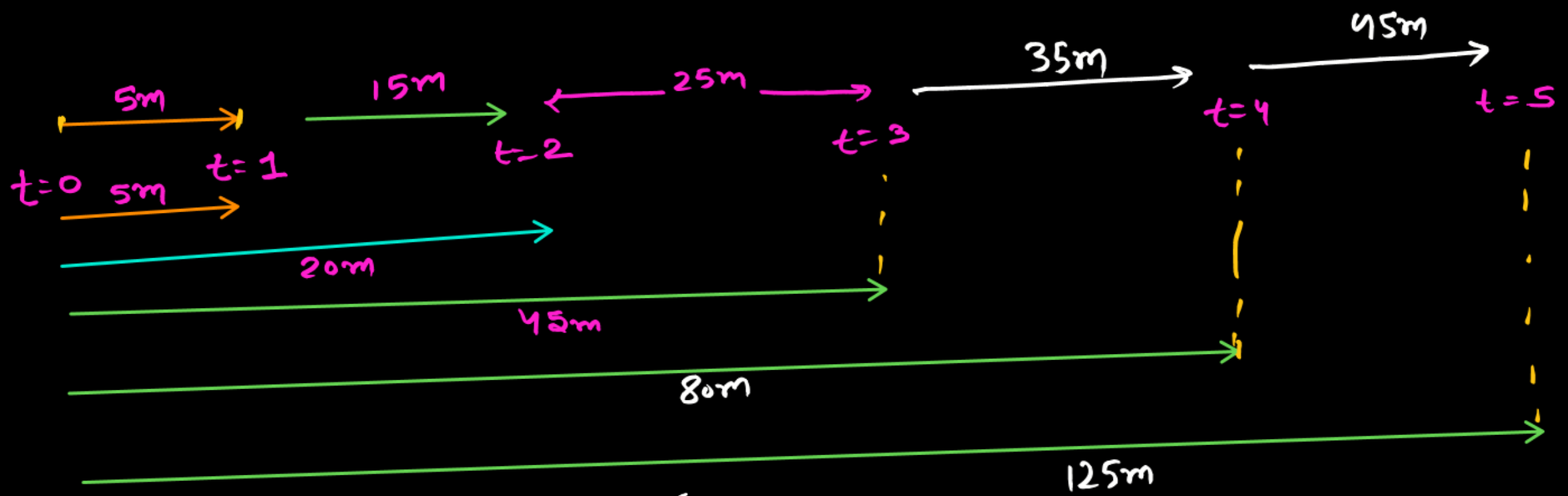
$= 1 : 3 : 5 : 7 : 9$

$= x : 3x : 5x : 7x : 9x$

$u=0$, $a=10m/s^2$, $t=6th\,n$

$u=0$, $a=10m/s^2$

IVEET



✓ Square karadi → $S_{1s} : S_{2s} : S_{3s} : S_{4s} = x : 4x : 9x : 16x$ ✓

✓ Odd no ka ratio → $S_{1st} : S_{2nd} : S_{3rd} : S_{4th} = x : 3x : 5x : 7x : 9x : 11x : 13x : 15x$

Copy (6) initial velocity of object is zero & const^n acc^n $= 4\,m/s^2$ then find disp^m in 2sec & disp^n in 4-sec & disp^n in 6sec

$$\begin{cases} S_{t=2s} = \frac{1}{2}(4)(2)^2 = 8m \\ S_{t=4sec} = \frac{1}{2}(4)(4)^2 = 32\,m \\ S_{t=6sec} = \frac{1}{2}(4)(6)^2 = 72\,m \end{cases}$$

$S_{2s} : S_{4s} : S_{6se} = 8 : 32 : 72$
$= 1 : 4 : 9$

$S_{(t)} : S_{(2t)} : S_{(3t)} = 1 : 4 : 9$ ✓✓

$a = const^n$
$u = 0$ (rest)

$S_{2sec} : S_{2sec-4sec} : S_{4sec-6se} = 8 : 24 : 40$
$= 1 : 3 : 5$ ✓

MP*

$S_{t\text{-}time} : S_{next} : S_{next\ to\ next} = 1 : 3 : 5$ ✓

$u = 0^{th}$ $a = const$ ✓

Question 7

Object starts his motion from rest and constant acceleration then find ratio of displacement in 6th sec and 6 sec.

$u=0$

$a=a$

$S_{nth} = u + \frac{a}{2}(2n-1)$

$$\frac{S_{6^{th}\,sec}}{S_{6\,sec}} = \frac{11}{36} = \frac{\frac{a}{2}(2n-1)}{\frac{1}{2}a(n)^2} = \frac{\frac{a}{2}(2\times6-1)}{\frac{a}{2}(6)^2} = \frac{11}{36}$$ ✓

Question

(8)

Object starts his motion from rest and constant acceleration and moves 100 m in 1st 5-sec then find displacement in next 5-sec.

$S_{0-5sec} : S_{5th-10} = n : 3n$

Imp*

$S_{5sec} = 100m \longrightarrow S_{next\ 5} = 3 \times 100 = 300m$ ✓



Physics

$u = 0 \quad t = 5 \quad S = 100m$

$S = ut + \frac{1}{2}at^2$

$100 = 0 + \frac{1}{2}(a)(5)^2$

$800 = a(25) \quad \boxed{a = 8}$

$S_{10sec} = \frac{1}{2}(8)(10)^2 = 400$

$S_{in\ 5\ to\ 10sec} = S_{10} - S_5$

$= 400 - 100$

$= 300m$ ✓

Question

A particle experiences a constant acceleration for 20 sec after starting from rest. If it travels a distance S_1 in the first 10 sec and a distance S_2 in the next 10 sec, then:

1. $S_1 = S_2$
2. $S_1 = S_2/3$
3. $S_1 = S_2/2$
4. $S_1 = S_2/4$

$S_1 : S_2 = 1 : 3$

$\frac{S_1}{S_2} = \frac{1}{3}$

$S_1 = \frac{S_2}{3}$

Question

(10) No Us

A motor car moving with a uniform speed of 20 m/sec comes to stop on the application of brakes after travelling a distance of 10 m. Its acceleration is:

$a = \frac{v_f - v_i}{\Delta t}$

1. $20\ \text{m/sec}^2$
2. $-20\ \text{m/sec}^2$
3. $-40\ \text{m/sec}^2$
4. $+2\ \text{m/sec}^2$

Question || No Copy

What will be the ratio of the distance moved by a freely falling body from rest in 4th and 5th seconds of journey? **[1989]**

1. 4 : 5
2. 7 : 9
3. 16 : 25
4. 1 : 1

$u = 0$ (rest)

$a = g$ ✓

$S_{n^{th}} = u + \frac{a}{2}(2n-1) = \frac{g}{2}(7)$

$\frac{g}{2}(9)$

No Copy

Question

A particle start moving from rest and constant acceleration. It travels a distance x in first $2s$ and y in $4s$ then correction relation is

1. $y = 3x$
2. $y = 4x$
3. $y = x$
4. $x = 4y$

Question

A body starts from rest travelled a distance 120 m in the 8th sec then acceleration is:

1. 10
2. 8
3. 16
4. 4

MRT Box

If time is given or asking then apply 1, 2, or 4th eqn of motion

$v = u + at$ $\quad s = ut + \frac{1}{2}at^2$

$S_n = u + \frac{a}{2}(2n-1)$

$S_m = u + \frac{a}{2}(2m-1)$

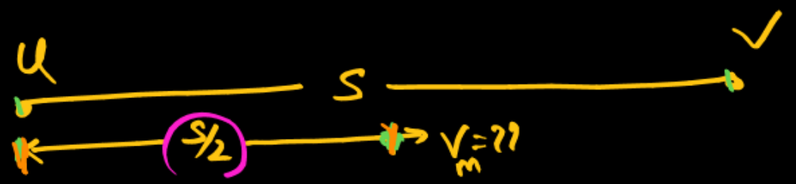
$120 = 0 + \frac{a}{2}(16-1)$

$8 \times 2 = a \times 1$

$a = 16\ m/s^2$

Copy

(8) object starts his motn with intial velocity u & cosn accn a aftr S dispm its velocity becon V then find velocity at mid of Journey.



Soln

3rd eqn of motn for compt Journey

$$V^2 - u^2 = 2aS \quad \text{---- (1)}$$

$$V_m^2 - u^2 = 2a\frac{S}{2} \quad \text{---- (2)}$$

$$\frac{V^2-u^2}{V_m^2-u^2} = 2$$

$$V^2-u^2 = 2V_m^2-2u^2$$

$$2V_m^2 = u^2+v^2$$

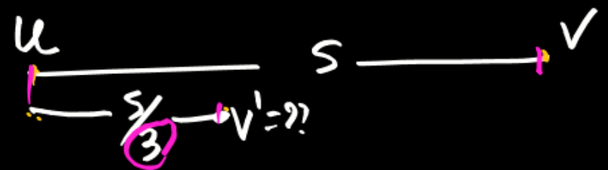
$$V_m = \sqrt{\frac{u^2+v^2}{2}}$$

40%
(a) required very rare
(b) No concept (60%)

$$V' = \sqrt{\frac{(n-1)u^2+v^2}{n}}$$

V' is velocity at disp = $\frac{S}{(n)}$

find velocity at one 3rd distn



$$V^2-u^2 = 2aS$$

$$V'^2-u^2 = \frac{2aS}{3}$$

$$\frac{V^2-u^2}{V'^2-u^2} = 3$$

$$3V'^2 = 2u^2+v^2$$

$$V' = \sqrt{\frac{2u^2+v^2}{3}}$$

Question

A small toy starts moving from the position of rest under a constant acceleration. If it travels a distance of 10m in t s, the distance travelled by the toy in the next t s will be:

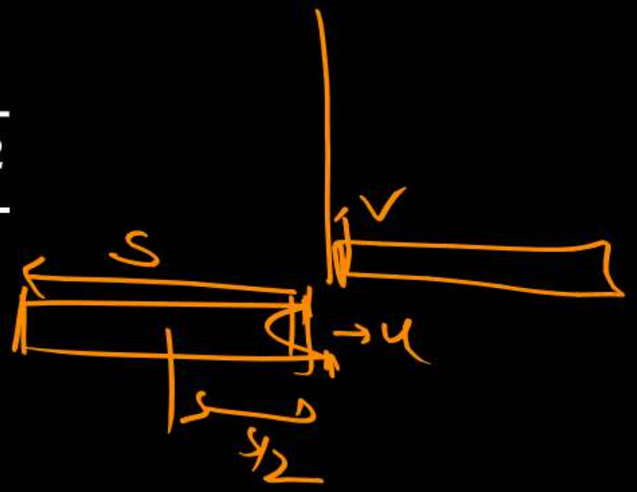
[JEE Main 2022]

1. 10 m
2. 20 m
3. 30 m
4. 40 m

Question

An engine of a train, moving with uniform acceleration, passes the signal-post with velocity u and the last compartment with velocity v. The velocity with which middle point of the train passes the signal post is: **[JEE Main 2021]**

1. $\frac{u+v}{2}$
2. $\sqrt{\frac{v^2+u^2}{2}}$
3. $\frac{v-u}{2}$
4. $\sqrt{\frac{v^2-u^2}{2}}$



Question

Copy

A car is moving along a straight road with a uniform acceleration. It passes through two points P and Q separated by a distance with velocity 30 km/h and 40 km/h respectively. The velocity of the car midway between P and Q is **[1988]**

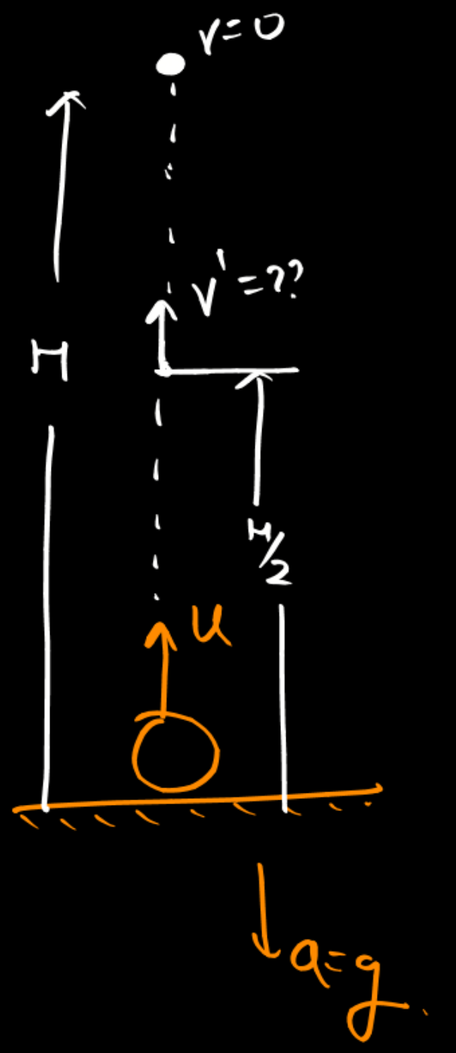
1. 33.3 km/h
2. $20\sqrt{2}$ km/h
3. $25\sqrt{2}$ km/h
4. 35 km/h

$$v_{mid} = \sqrt{\frac{30^2 + 40^2}{2}}$$

$$= \frac{50}{\sqrt{2}} = 25\sqrt{2}$$

Copy

(Q) Ball is Project up with speed u
find its velocity it half of maximum height.



$$u' = \sqrt{\frac{u^2+0}{2}} = \frac{u}{\sqrt{2}}$$ ✓

(Q) Object starts his motion from rest & constant acceleration a, takes time T to cover S displacement. then find time in 1st & 2nd half distance



$t_1 = t_2$ ✗

$t_1 < t_2$ ✗

$t_1 > t_2$ ✓

Sol^n for compt Journey

$$S = ut + \frac{1}{2}at^2$$

$$S = \frac{1}{2}aT^2 \quad — ①$$

for 1st half Journey

$$\frac{S}{2} = 0 + \frac{1}{2}at_1^2$$

$$\frac{S}{2} = \frac{1}{2}at_1^2 \quad — ②$$

② / ①

$$\frac{\frac{S}{2}}{S} = \frac{\frac{1}{2}at_1^2}{\frac{1}{2}aT^2}$$

$$\frac{T^2}{2} = t_1^2$$

$$t_1 = \frac{T}{\sqrt{2}}$$

for 2nd half

$$S = ut + \frac{1}{2}at^2$$

$$\frac{S}{2} = 0 + \frac{1}{2}at_2^2$$

$$t_2 = \frac{T}{\sqrt{2}}$$ ✗

$$t_1 + t_2 = T$$

$$t_2 = T - t_1 = T - \frac{T}{\sqrt{2}} = \frac{\sqrt{2}T - T}{\sqrt{2}}$$

$$t_2 = \frac{(\sqrt{2}-1)T}{\sqrt{2}}$$

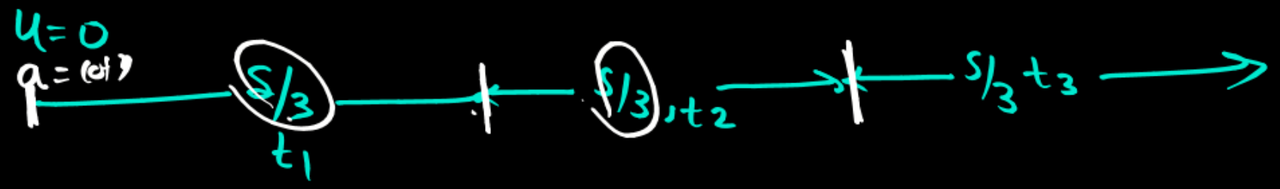
Ratio of time to cover equal distn $s/2$ starting from rest

$\sqrt{1} = 1$

$$\frac{t_1}{t_2} = \frac{\cancel{I/\sqrt{2}}}{\cancel{\frac{I}{\sqrt{2}}}(\sqrt{2}-1)}$$

$$t_1 : t_2 = 1 : \sqrt{2}-1$$

#

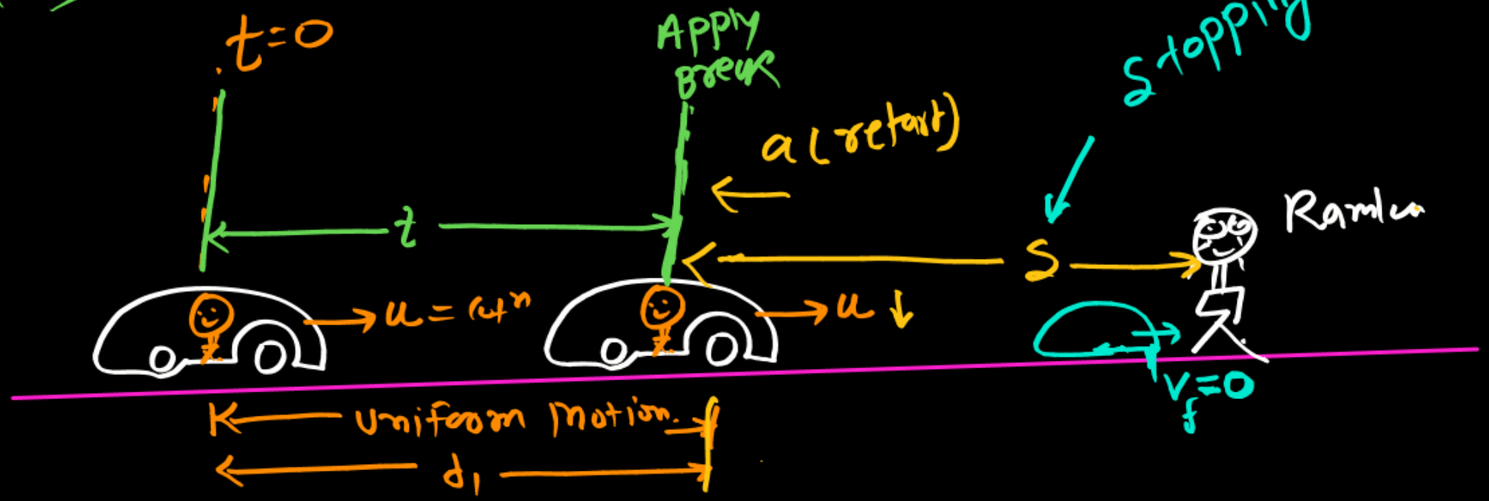


$$t_1 : t_2 : t_3 = \sqrt{1}-\sqrt{0} : \sqrt{2}-\sqrt{1} : \sqrt{3}-\sqrt{2} : \sqrt{4}-\sqrt{3}$$

$$= 1 : \sqrt{2}-1 : \sqrt{3}-\sqrt{2}$$

Reaction time →

think → I have to apply break.



$d = u \times t$

$\left\{ t = \frac{d}{u} \right\}$

$S \propto (u)^2$ *

3rd eqn of motn

$v_f^2 - u_i^2 = 2as$

$0 + u^2 = 2(a)(S)$

$S = \left[\frac{u^2}{2a} \right]$

$\left\{ \begin{array}{l} S \propto (u)^2 \\ S \propto \frac{1}{a} \end{array} \right]$

Question

A bus moving along a straight highway with speed of 72 km/h is brought to halt within 4s after applying the brakes. The distance travelled by the bus during this time (Assume the retardation is uniform) is _____ m. **[JEE Main 2024]**

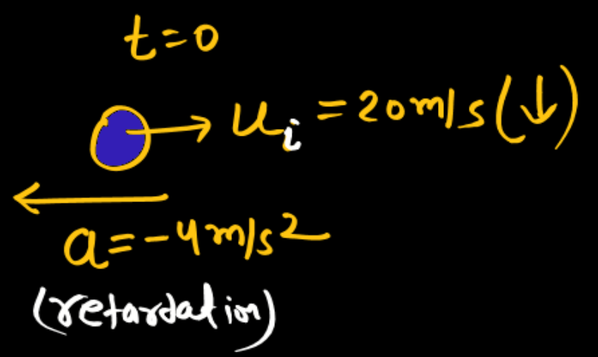
Question

A bus moving along a straight highway with speed of 72 km/h is brought to halt within 4s after applying the brakes. The distance travelled by the bus during this time (Assume the retardation is uniform) is _____ m. **[JEE Main 2024]**

Question

Object starts his motion with u and constant acceleration a then find its velocity at one 3rd displacement of complete journey if final velocity is V.

Find distance travelled by object in 6-sec.



Soln

displacement

$\# S = ut + \frac{1}{2}at^2$

$= 20 \times 6 + \frac{1}{2}(-4)(6)^2$

$= 120 - \frac{1}{2} \times 4 \times 36$

MRScam (-1)

$= 120 - 72$

$\boxed{S = 48 m}$

Wrong

displacement in 6 sec = 48 m ✓

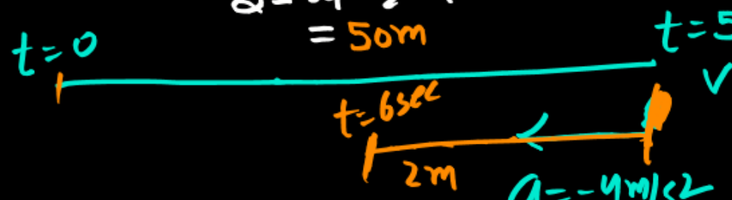
* time when object is at rest (v = 0)

$v = u + at$

$0 = 20 - 4t$

$t = 5 sec$

$u = 4 m/s$ ✓



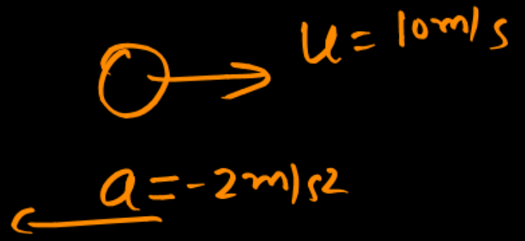
$disp^n = 50 + 2 = 52$

$disp = 50 - 2 = 48 m$

$S = ut + \frac{1}{2}at^2$

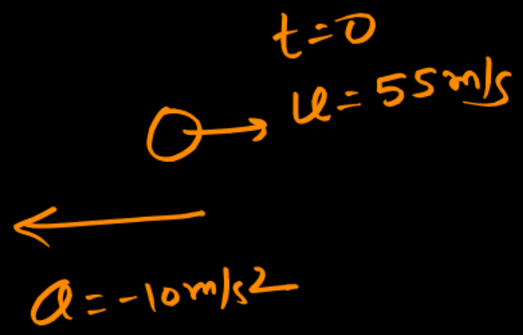
$= 0 + \frac{1}{2}(-4)(1)^2$

$= -2 m$



dist^n in 5-sec. | dist^n in 8-sec

Q



find disⁿ in 6-sec & disⁿ in 6th sec (5sec to 6sec)



Join it

Thank you